Usability test plans, recruitment and reporting

Usability test plans

- **Usability test plans** document all aspects of how the test will occur, including:
  - Scope
  - Purpose
  - Schedule / location
  - Format
  - Participants
  - Equipment
  - Roles
  - Sessions
  - Tasks (or Scenarios)
  - Observations
  - Metrics
  - Instruments (Surveys, Interviews)
  - Participant instructions (in the case of asynchronous testing)

Let's go over each in more detail...

Each of these could be a section in a usability test plan.

The questions are what information the test plan could provide.

Scope

- What application is being tested?
  - e.g. Microsoft Teams
- What feature(s) of the application are being tested?
  - e.g. the new breakout room feature
- When is the testing occurring? What version of the product or interface?
  - February 2021?
  - Version 10.5 of the application
  - Sketches? Interactive prototype?

Purpose

- What are you concerned with? What questions do you want to answer through the usability test?
- What are your goals in conducting the testing?
  - Goals can be very general or very specific
- General goals:
  - Can users easily learn how to use the application?
  - Do users commit many errors when using the app?
- Specific goals:
  - Can users use the shopping cart successfully?
  - How long will it take users to find the advanced settings?

Goals

- You only have so much time with each user
  - How many goals can you realistically achieve?
  - Don't have too many goals
- Goals should drive decisions about the design of the rest of the test plan
- Goals should be carefully throughtout, tied to goals for your application and concerns about its design

Schedule and location

- When and where will your tests take place?
  - Physical location? Could be "in the field", in a lab, in a coffee shop, etc.
  - Date and time?
- Will it be done remotely or in-person?
- Be specific about how many sessions you will hold

Format

- Will it be moderated or unmoderated?
  - i.e. synchronous vs. asynchronous

- **Synchronous** usability tests are moderated, with tester guiding the user verbally through the tasks, directly interviewing and asking follow-up questions

- **Asynchronous** usability tests are unmoderated, with tester giving the participant a script that they follow themselves
  - Follow-up questions are still possible over e-mail, chat, etc.

Participants

- Who are the participants in your usability test?
  - Do you know who they are?
  - Do you know any demographic information?
- How were the participants recruited, or how are they going to be recruited?
  - Advertisements, e-mails to newsletters, etc.
- How well do the recruited participants match your target demographic(s)?

Equipment

- What equipment will be used during the test?
  - Hardware?
  - Software?
  - Recording devices?
- What will be recorded with equipment, if anything?
- If it matters, we may want to describe screen resolutions, screen sizes, input devices, etc.

Roles

- Who will conduct the usability tests, and what role will they play?
  - It may be one person, but when resources allow for it, it might be a team conducting tests
  - A team of people might conduct one test with each participant, or a team may test with different participants in parallel
- A usability specialist might conduct the test, and a notetaker might take primary responsibility for taking notes
  - Other members of the team, including developers, may be invited as observers and also take notes

Sessions

- What is the order of activities for a test session with a participant? Example:
  1. Pre-study survey
  2. Ask participant to perform task X
  3. Ask participant to perform task Y
  4. Ask participant to perform task Z
  5. Post-study survey
  6. Post-study interview

- How long will a session take?
  - 60-90 minutes is typical in testing a large application
  - Tests can be done in 10-20 minutes for smaller applications or for tests focused on particular aspects of an application

Tasks and scenarios

- What tasks will you ask users to carry out?
  - Tasks can come from a task hierarchy, at different depths in the tree
- Tasks are generally better written in the form of scenarios with specific information
  - These might be called "task scenarios" as well
  - e.g. instead of "use the app to buy a book" the task will be "use the app to find and buy the book 1984"
  - i.e. ask the user to do something specific rather than the general task description itself

Tasks and scenarios

- **Exploratory tasks** are open-ended and may not have a specific correct answer
  - Useful for learning how people discover information, not for quantitative results
  - e.g. "You are interested in booking a vacation for your family. See if the site offers anything that you might suit your needs."

- **Specific tasks** are more focused and have a definitive end point
  - e.g. Find the Saturday opening hours for the Sunnyvale public library
  - Can be used to gather both quantitative (e.g. task performance time) and qualitative data

Observations

- How will you be making observations?
  - Recording via cameras, microphones, eye-tracking?
  - Screen recording?
  - Note taking?

- Metrics and instruments
  - Will you be recording any metrics?
  - Will you be conducting any interviews?
  - Will you be conducting any surveys?

Metrics

- What quantitative metrics will you measure?
  - e.g. task performance time in seconds
  - e.g. number of errors for a task
  - e.g. successful task completion rates

- How will you measure them?
  - e.g. stopwatch
  - e.g. note-taking each time a user makes an error
  - e.g. in-app logging

Metrics

- Will you produce metrics for user feelings and thoughts that don't have straightforward quantifications?
  - e.g. subjective satisfaction
  - e.g. trustability
- How will you measure them?
  - e.g. survey instruments (perhaps with likert scales)
  - e.g. analysis/summarization of interview responses
  - Will they be measured per task, per test, or both?

	The Post-Study Usability Questionnaire Version 3		Strongly agree 1	2	3	4	5	6	Strongly disagree 7		NA
1	Overall, I am satisfied with how easy it is to use this system.		O	O	O	O	O	O	O		O
2	It was simple to use this system.		O	O	O	O	O	O	O		O
3	I was able to complete the tasks and scenarios quickly using this system.		O	O	O	O	O	O	O		O
4	I felt comfortable using this system.		O	O	O	O	O	O	O		O
5	It was easy to learn to use this system.		O	O	O	O	O	O	O		O
6	I believe I could become productive quickly using this system.		O	O	O	O	O	O	O		O
7	The system gave error messages that clearly told me how to fix problems.		O	O	O	O	O	O	O		O
8	Whenever I made a mistake using the system, I could recover easily and quickly.		O	O	O	O	O	O	O		O
9	The information (such as online help, on-screen messages and other documentation) provided with this system was clear.		O	O	O	O	O	O	O		O
10	It was easy to find the information I needed.		O	O	O	O	O	O	O		O
11	The information was effective in helping me complete the tasks and scenarios.		O	O	O	O	O	O	O		O
12	The organization of information on the system screens was clear.		O	O	O	O	O	O	O		O
13	The interface* of this system was pleasant.		O	O	O	O	O	O	O		O
14	I liked using the interface of this system.		O	O	O	O	O	O	O		O
15	This system has all the functions and capabilities I expect it to have.		O	O	O	O	O	O	O		O
16	Overall, I am satisfied with this system.		O	O	O	O	O	O	O		O

**The "interface" includes those items that you use to interact with the system. For example, some components of the interface are the keyboard, the mouse, the microphone, and the screens (including their graphics and language).*

Source: https://i.pinimg.com/originals/df/3a/fb/df3afb126cda3ce4b8570d79c1550d7c.png

Surveys

- Will participants complete a survey(s)?
- What surveys will participants complete?
- What questions are on the surveys?
  - Are standard surveys like SUS being used?
- When will participants complete the surveys?
  - Pre-test? Post-test?

Interviews

- Will interviews be conducted?
  - When will the interviews be conducted?
- Is there a script of preset interview questions?
  - Are there potential follow-up questions ready?
- Will the interview by unscripted, or will the interview contain unscripted questions?

Interview alternatives

- If in the case of an asynchronous / unmoderated remote usability test interviews are not possible, can anything be done as an alternative?
  - E-mail? Chat messages?

- If so, the same or similar information that would be provided for an interview can also be provided
  - But also state how you're planning on following up afterwards

Instructions

- If the usability test is asynchronous (aka unmoderated), the test plan should include instructions for the tester to complete the test, including:
  - How to access the application
  - Any pre-test surveys (e.g. link to a Google Form)
  - Is screen and mic recording required during the testing?
    - How to send the recording to the tester afterwards?
  - What tasks to attempt (same description as in your test plan)
    - Do you want the participant to think-aloud into a microphone as they perform each task?
  - Any post-test surveys
  - Any post-test chat or e-mail follow-up conversations

- Tester can be expected to carry out instructions

Usability test plans

- This is an overview in full potential detail
  - Not all of these sections are relevant for all usability tests
  - Not all of the questions need answers, maybe they aren't relevant for your test
  - Some of these "sections" can be very short, a sentence or a few point form comments

- What's important is designing the test in order to learn, and the formal documentation of that design

Usability test plan templates

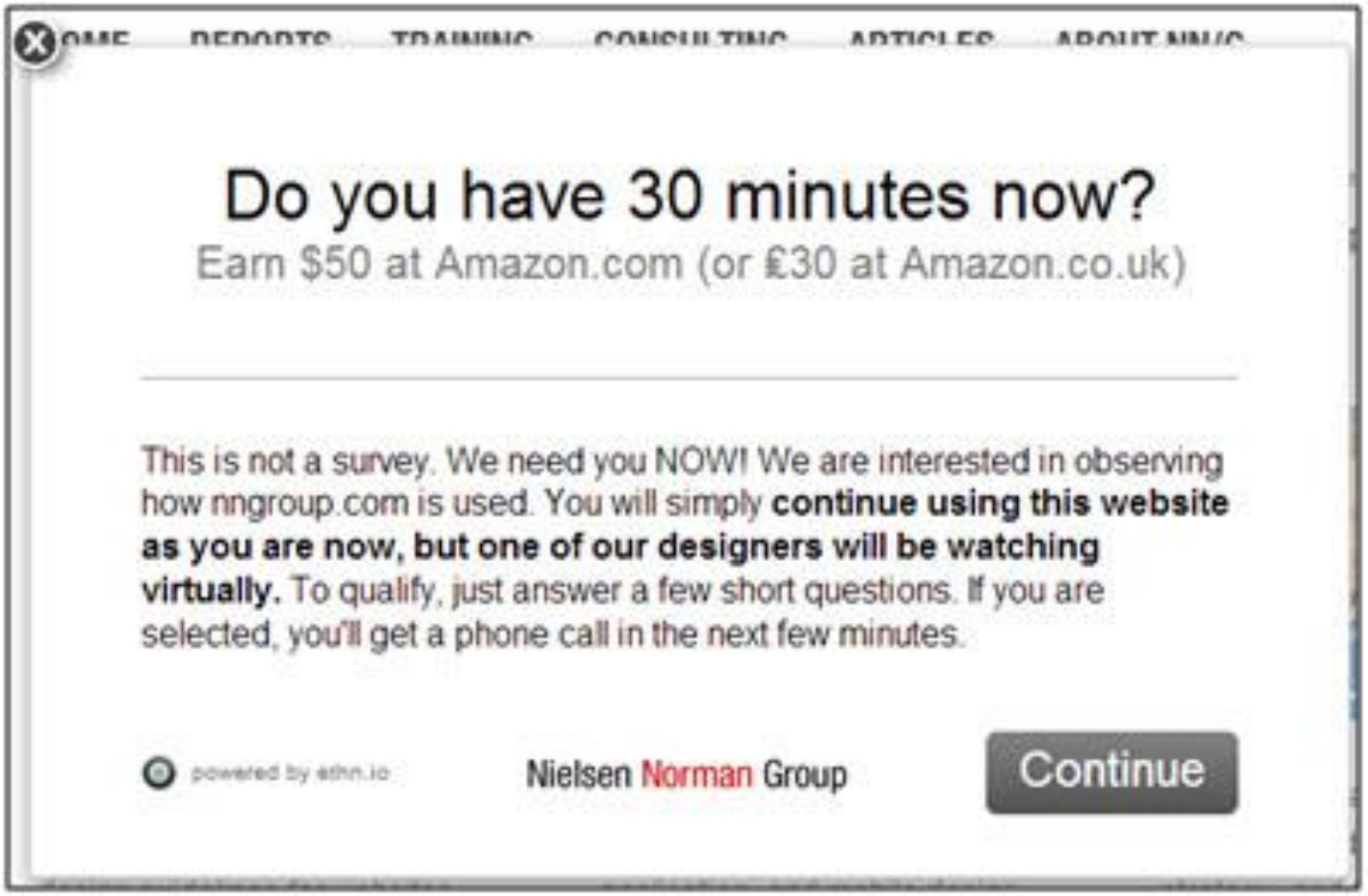
- Templates exist for usability test plans
  - https://www.usability.gov/how-to-and-tools/resources/templates/usability-test-plan-template.html
  - https://www.shanedoyle.io/post/a-template-for-usability-testing
  - 12 templates, including for specific situations: https://www.hotjar.com/usability-testing/template-checklist/
- Templates can save you time and help make sure you're including everything required
  - But they aren't purpose-fit for your application and test so be cautious not to depend on them too heavily... great for ideas!

Usability test plan examples

- Examples:
  - Centre for Social Innovation website test plan - http://danielleklein.ca/wp-content/uploads/2016/08/Usability-test-plan-CSI.pdf

- Examples of task scenarios:
  - https://blog.testlodge.com/usability-testing-examples/

- A "lean" usability one page test plan
  - https://medium.com/@userfocus/the-1-page-usability-test-plan-dbc8c3d7fb54



Source: https://s3.amazonaws.com/media.nngroup.com/media/editor/2013/12/03/1_screener_p1.jpg

Recruiting participants

- The most important thing is to ensure participants are similar to your target user!
  - If you have multiple groups of target users, recruit participants similar to each group
  - Beyond this the actual form of recruitment can be whatever works... e-mails, social media posts, posters, etc.

- Avoid conflicts of interest
  - Internal staff or people who have interests in the project may not be as honest as external participants

Recruiting participants

- Discount usability testing suggests 5 participants

- If we want to produce a statistically significant result, at least 20 participants
  - Though often more are required
  - Tests like this can begin to resemble formal HCI experiments done by HCI researchers
  - At this point, ensuring that the recruitment is "random" is also important

Recruiting participants

- We can screen participants...
  - Demographics (age, gender, etc.)
  - Personality
  - Work experience
  - Background knowledge (including education)
- For niche applications, screening becomes more important
- Compensating participants can help ensure they take the test more seriously

Usability test report

- Usability test results might be reported verbally in a meeting, perhaps with slides, or might be reported in a formal written report

- Usability test reports should include:
  - Background summary
  - Methodology
  - Test results
  - Analysis of results
  - Recommendations

- We can create sections with these headings, but like usability test plans, we don't have to use these exactly
  - We may have additional sections, e.g. "Executive Summary"

Background summary

- Background summary and methodology are both made up of essentially the same information as the test plan
  - But the audience is potentially different... testers vs. managers for example
  - Information is generally best summarized
  - Usability test report should be more universally accessible

- Made up of these portions of the test plan, summarized for the reader:
  - Scope
  - Purpose
  - Schedule / location

Methodology

- Made up of these portions of the test plan, summarized for the reader:
  - Format
  - Participants
  - Equipment
  - Roles
  - Sessions
  - Tasks (or Scenarios)
  - Observations
  - Metrics
  - Instruments (Surveys, Interviews)
  - Participant instructions

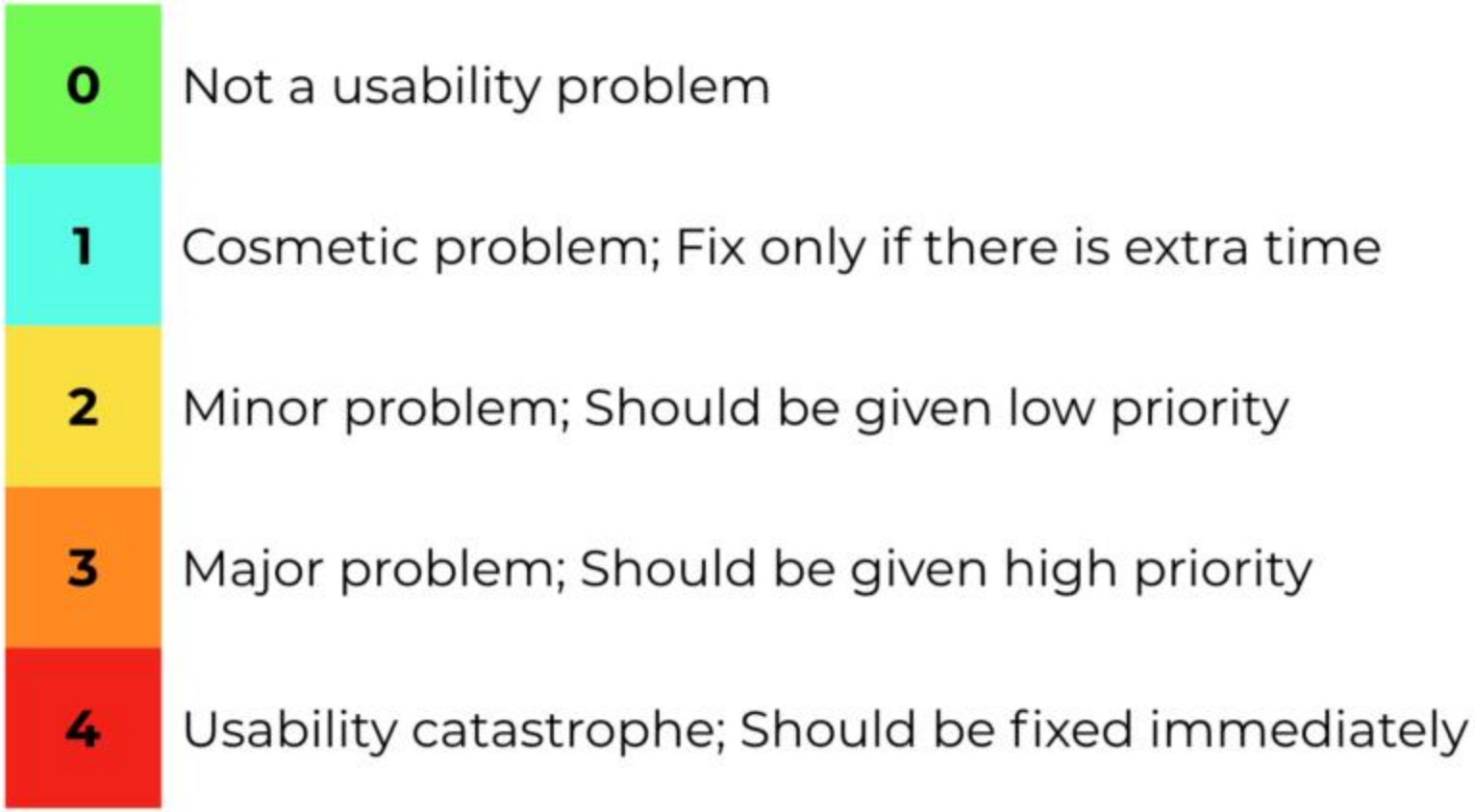
Test results

- Present the data that was collected:
  - Survey data
  - Metrics recorded by the application
  - Interview data
  - Note-taking observations
- Utilize tables, graphs, screen captures wherever illustrative or helpful to present the data and results
  - e.g. test of performance times
  - e.g. where in the app an issue is discussed
- Beyond what you intended to measure and observe, was there anything unexpected to report?
  - e.g. programming bugs

Analysis

- Analyze the results should answer: what was learned? What is evident or likely from the results?
  - We can call what was learned **findings**
- How does the data support the findings?
- What usability issues were identified?
  - Can organize them by priority, for example:
    - Low – mostly cosmetic issues
    - Medium – can be frustrating for users
    - High – must be fixed as soon as possible
    - Critical – app cannot be released without fixing them
  - Frequency of occurrence of the issue may be reported, and may be a factor in the issue's severity level

Can use color coding in the report



Source: https://www.clarencechoy.com/wp-content/uploads/2019/10/Screen-Shot-2019-10-22-at-2.40.28-PM-768x424.png

Analysis

- Findings can be reported per-scenario, aggregate across scenarios, or both
- Analysis should not be exclusively negative
  - What positives were found?
  - What is working well should be continued!
  - Good for the morale of the development team
- Findings should be as specific as possible

FINDINGS SUMMARY TABLE

ISSUE TYPE	ISSUES	PAGE	RECOMMENDATIONS	EASE OF FIXNG
Button Missing	The EmbodyDNA icon was clicked on for navigating to the home page.	Embody DNA	Make the “embodyDNA” page icon lead to the home page.	Easy
Button Missing	Users ended up choosing "Lose it!" Button to go to the home page.	Embody DNA	Make the “embodyDNA” icon lead to the home page.	Easy
Password Verification	Users have recommended adding an interactive confirmation for matching passwords.	Create New Account	Add an interactive password verification checkbox.	Difficult
Measurement	Several users would like to verify the exact weight measurement.	Create New Account	Add the actual metric values next to US and Metric.	Medium
Add New Button	“Create New Account” button is not provided.	Home	Add a "Create New Account Button" next to the "Log in" button.	Easy
Naming Convention	"Get Lose it!" button was constantly pressed for creating a new account.	Home	Change the button title to say "Down Load Mobile App Here."	Easy
Looping Page	"Continue to loseit.com" button was clicked looping the user back to the home page.	Download App	Change the button title to say "Create New Account."	Easy

Source: https://uploads-ssl.webflow.com/57ae10c2ec62b90517b4a868/5bb45248a4b92393b984b795_Page_30%402x.png

Recommendations

- **Recommendations** are the changes recommended as a result of the findings of the usability test
- Recommendations should be associated with findings
- What is the reasoning for the recommendation?
  - This is not the same as the findings
  - The reasoning is why you believe the recommendation will address the finding
  - e.g. "we should move the search bar to the top-right, because that's where users were moving their focus to look for it"

Usability test report examples

- Weight loss app usability test report:
  - https://uploads-ssl.webflow.com/57ae10c2ec62b90517b4a868/5ba010ffc492f6fd7086659b_Usability_Testing_Presentation_Report.pdf
  - Notice the accessible nature of the document, as opposed to a more verbose technical document

- U-haul.com usability test report:
  - https://uxqb.org/wp-content/uploads/documents/CPUX-UT_EN_Usability-Test-Report-Example.pdf
  - Note how his report is more formal

- Digital health system usability report:
  - https://www.drummondgroup.com/pdfs/MEDHOST-Usability-Report-Enterprise-2017R1-1.pdf

More resources

- Usability testing basics:
  - http://webservices.itcs.umich.edu/drupal/wwwsig/sites/webservices.itcs.umich.edu.drupal.wwwsig/files/Usability-Testing-Basics.pdf

- Usability testing:
  - https://www.usability.gov/how-to-and-tools/methods/usability-testing.html